

REPORT

ARTIFICIAL INTELLIGENCE

Assessment – 1: Analytical Problem Solving

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Course: Artificial Intelligence

Assessment Title: Analytical Problem Solving

1. Introduction

Artificial Intelligence (AI) is a branch of computer science that focuses on designing intelligent systems capable of performing tasks that normally require human intelligence. AI techniques enable machines to solve problems, make decisions, learn from experience, and interact with their environment. Analytical problem solving is one of the core areas of AI, where search algorithms and intelligent agent models are used to obtain optimal solutions.

This assessment includes five important AI problems: the Water Jug Problem, Mars Rover Intelligent Agent, 8-Queens Problem, OLA Cab Booking Intelligent Agent, and Uniform Cost Search (UCS). Each problem demonstrates different AI concepts such as state-space search, intelligent agents, optimization, and search strategies.

2. Objective

The main objectives of this assessment are:

- To understand the concept of Artificial Intelligence problem solving.
- To study different intelligent agent models.
- To apply search algorithms for solving AI problems.
- To understand state-space representation and search techniques.
- To implement AI concepts using Python programming.
- To analyze real-world applications of intelligent agents.

3. Software Requirements

- Python 3.x
- Notepad / Visual Studio Code
- GitHub
- Windows Operating System

4. Problems Implemented

Problem 1 – Water Jug Problem

The Water Jug Problem is a classical Artificial Intelligence search problem. Two jugs of capacities 4 gallons and 3 gallons are provided without any measurement markings. Using only filling, emptying, and transferring operations, exactly 2 gallons of water must be obtained in the 4-gallon jug.

This problem demonstrates state-space search and systematic problem solving.

Problem 2 – Mars Rover Intelligent Agent

The Mars Rover is an autonomous intelligent agent that explores the surface of Mars. It collects scientific samples, avoids obstacles, monitors battery status, and communicates with Earth.

The PEAS model was identified as:

- **Performance Measure:** Safe navigation, successful sample collection, battery efficiency.
- **Environment:** Mars surface with rocks, sand, hills, and craters.
- **Actuators:** Wheels, robotic arm, drill, communication system.
- **Sensors:** Camera, proximity sensor, battery sensor, temperature sensor.

The suitable AI model is the **Learning Agent**, since it adapts to changing environmental conditions.

Problem 3 – 8-Queens Problem

The 8-Queens Problem is a constraint satisfaction problem where eight queens are placed on a chessboard such that no queen attacks another queen.

The Backtracking Search Algorithm was used to generate a valid solution. This problem demonstrates state-space search and recursive problem-solving techniques.

Problem 4 – OLA Cab Booking Intelligent Agent

The OLA Cab Booking System is modeled as a Goal-Based Intelligent Agent. The system accepts the customer's source and destination, displays available cab types, estimates the fare, and confirms the booking.

The agent selects the most suitable cab to satisfy the customer's travel requirements efficiently.

Problem 5 – Uniform Cost Search (UCS)

Uniform Cost Search is an uninformed search algorithm that always expands the node with the lowest cumulative path cost.

Using the given weighted graph, UCS successfully identifies the least-cost path from the start node **S** to the goal node **G**, ensuring an optimal solution.

5. Algorithms Used

- Water Jug Search Strategy
- Learning Agent Model
- Backtracking Search Algorithm
- Goal-Based Agent
- Uniform Cost Search (UCS)

6. Python Implementation

Python programs were developed for all five problems. Each program demonstrates the corresponding Artificial Intelligence concept and produces the required output.

The implementation includes:

- Water Jug simulation
- Mars Rover agent representation
- 8-Queens solution using Backtracking
- OLA Cab Booking simulation
- Uniform Cost Search implementation

7. Applications

The concepts implemented in this assessment have many real-world applications:

- Robot Navigation
- Autonomous Space Exploration
- Smart Transportation Systems
- Route Optimization

- Game Playing
- Logistics and Supply Chain Management
- Decision Support Systems

8. Advantages

- Improves logical thinking and analytical skills.
- Demonstrates practical AI concepts.
- Helps understand intelligent agents.
- Provides knowledge of search algorithms.
- Introduces optimization techniques.
- Enhances Python programming skills.

9. Result

All five analytical problems were successfully implemented and analyzed using appropriate Artificial Intelligence models and search algorithms. The Python programs executed successfully and generated the expected outputs. The objectives of the assessment were achieved.

10. Conclusion

This assessment provided practical experience in solving Artificial Intelligence problems using intelligent agent models and search algorithms. Classical AI problems such as the Water Jug Problem, Mars Rover Agent, 8-Queens Problem, OLA Cab Booking System, and Uniform Cost Search were successfully studied and implemented in Python. The assessment improved understanding of state-space search, optimization, and intelligent decision-making techniques, demonstrating how AI methods can be applied to solve real-world problems efficiently.